

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE422: Artificial Intelligence Lab

Project Report

Diabetes Detection Using Machine Learning

A Supervised and Unsupervised Learning Approach

Submitted by

Kamrul Hassan Niloy

Student ID: 23101342

Shafaitul Kabir Rudro

Student ID: 23201605

Fall 202

Table of Contents

1. Introduction.....	1
2. Dataset Description.....	1
3. Dataset Pre-processing.....	8
4. Dataset Splitting.....	10
5. Model Training & Testing.....	10
6. Model Selection / Comparison Analysis.....	13
7. Conclusion.....	18

Introduction

In this project, a machine learning model is to be developed and evaluated with regard to its accuracy in prediction of whether an individual is diabetic based on demographic and medical factors such as age, body mass index, HbA1c level, glucose level, hypertension, heart disease, and smoking status. The performance of four machine learning models, namely Decision Tree, K-nearest neighbors (KNN), logistic regression, and neural network is evaluated in diabetes classification. Moreover, K-means clustering technique is considered to reveal inherent clusters within the patients.

Dataset Description

The dataset used in this study involves patient data with regard to demographics and clinical information for predicting diabetes, where diabetes is referred to as the target variable.

Overview

Property	Value
Number of data points (rows)	100,000
Number of features (columns)	9 (8 input features + 1 target)
Problem type	Classification (binary)
Target variable	diabetes (0 = No, 1 = Yes)

It is a classification problem and not a regression problem due to the nature of the target variable as discrete in nature (it is either 0 or 1). Therefore, the main aim is to predict the class label, which is the very hallmark of a classification problem.

Feature Types

The dataset contains a mix of categorical and quantitative (numerical) features:

- Categorical features: gender (Female / Male / Other), smoking_history (never, former, current, not current, ever, No Info).
- Quantitative features: age, hypertension (binary 0/1), heart_disease (binary 0/1), bmi, HbA1c_level, blood_glucose_level.
- Target: diabetes (binary 0/1).

Since gender and smoking history are categorical variables, they cannot be used as inputs into any ML algorithm because ML algorithms can take only numeric inputs. Therefore, these categorical variables had to be converted to numeric values before the use of an algorithm (refer Section 3 for further details).

Correlation Analysis

Correlation heat map has been generated for all the numeric features, including the target feature using the seaborn package to understand linear relationships between the variables

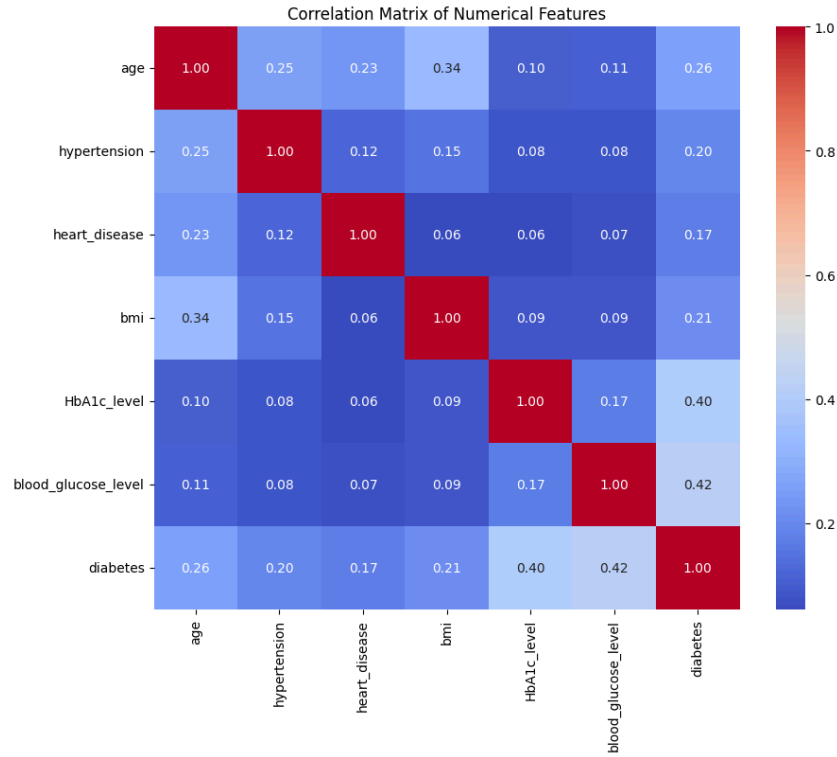


Figure 2.1: Correlation heatmap of numerical features.

From the correlation analysis, we find that there is a very strong positive correlation between blood glucose level ($r \approx 0.42$) and HbA1c level ($r \approx 0.40$) and diabetes, as per the existing medical literature because these two are the direct indicators of glucose control. There is a moderate positive correlation between age ($r \approx 0.26$), body mass index (BMI) ($r \approx 0.21$), hypertension ($r \approx 0.20$), and heart disease ($r \approx 0.17$) and diabetes. There is no particular high correlation among any input features.

Class Imbalance

There is significant class imbalance in the target attribute since there are 82,284 cases in the dataset belonging to the non-diabetic class and 7,634 cases that belong to the diabetic class, resulting in a 91% to 9% distribution.

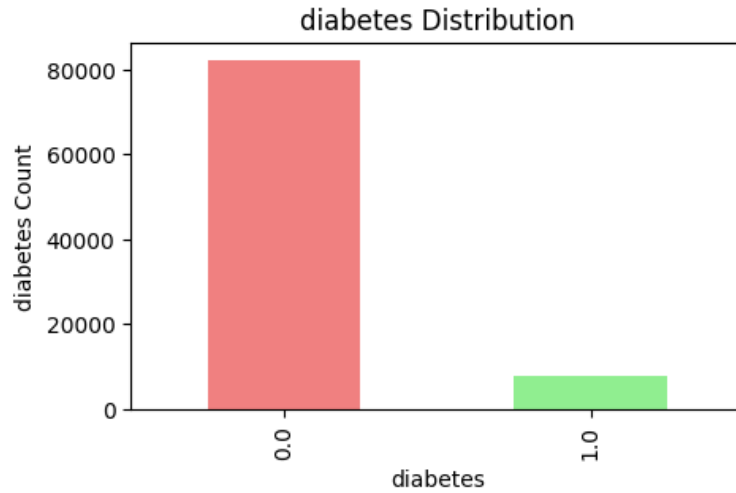


Figure 2.2: Class distribution of the diabetes target feature.

It is essential that this bias is taken into account while evaluating the performance of the model, since even if the model shows a very high level of accuracy through its predictions, it may just be because it is classifying the majority of non-diabetics, as shown in Section 6.

Exploratory Data Analysis

Some visualizations have been created to explore the relationship between each feature and the diabetic condition.

Demographic and lifestyle distributions

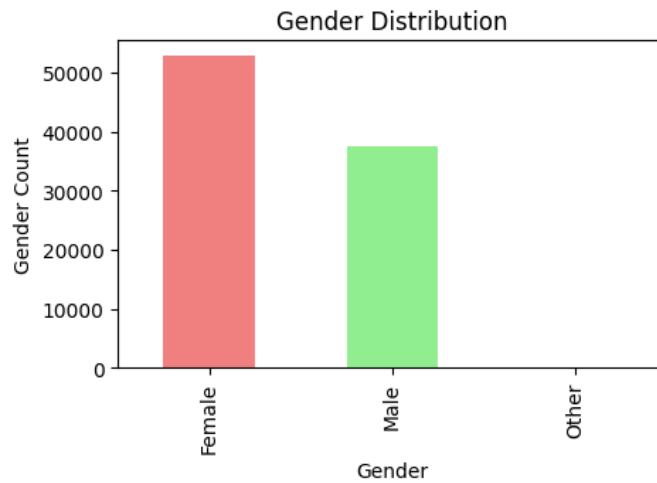


Figure 2.3: Gender distribution in the dataset.

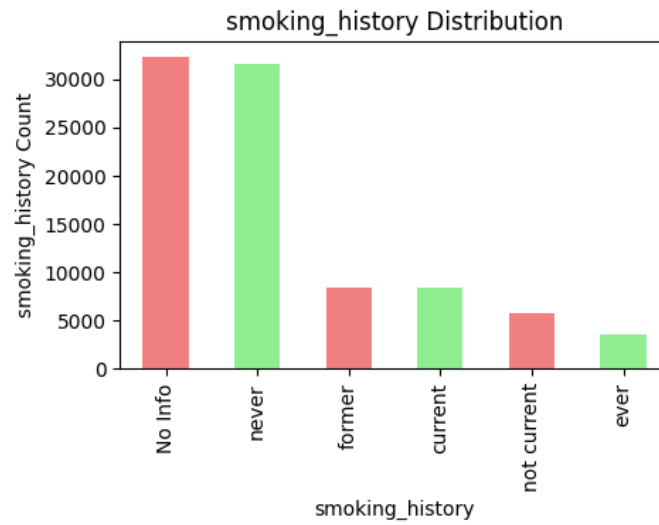


Figure 2.4: Smoking history distribution.

Diabetes incidence by categorical/binary features

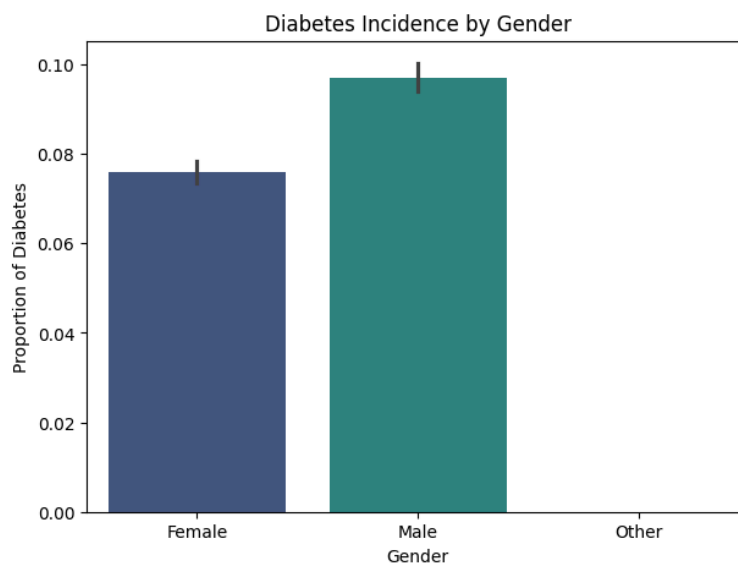


Figure 2.5: Diabetes incidence by gender.

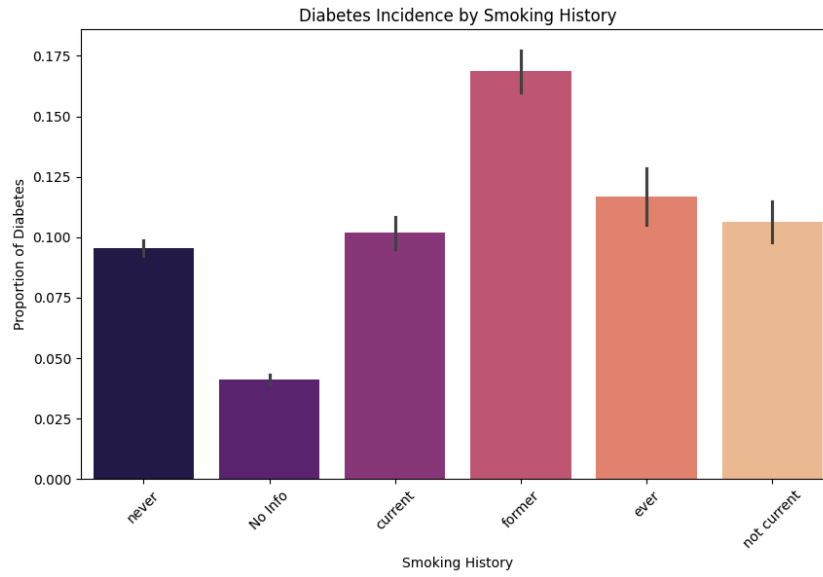


Figure 2.6: Diabetes incidence by smoking history.

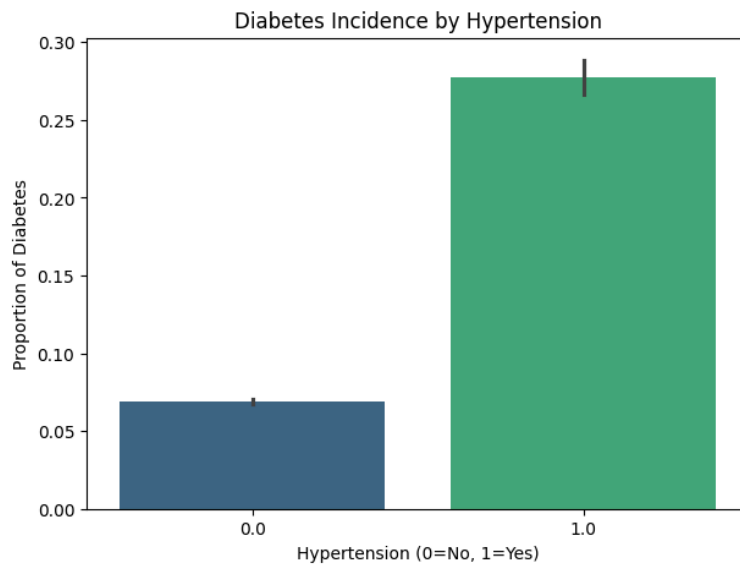


Figure 2.7: Diabetes incidence by hypertension status.

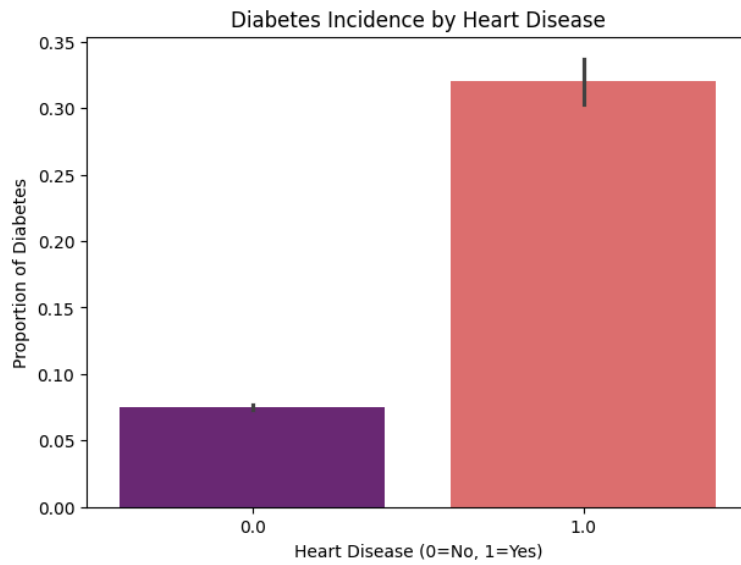


Figure 2.8: Diabetes incidence by heart disease status.

Diabetes incidence by numerical features

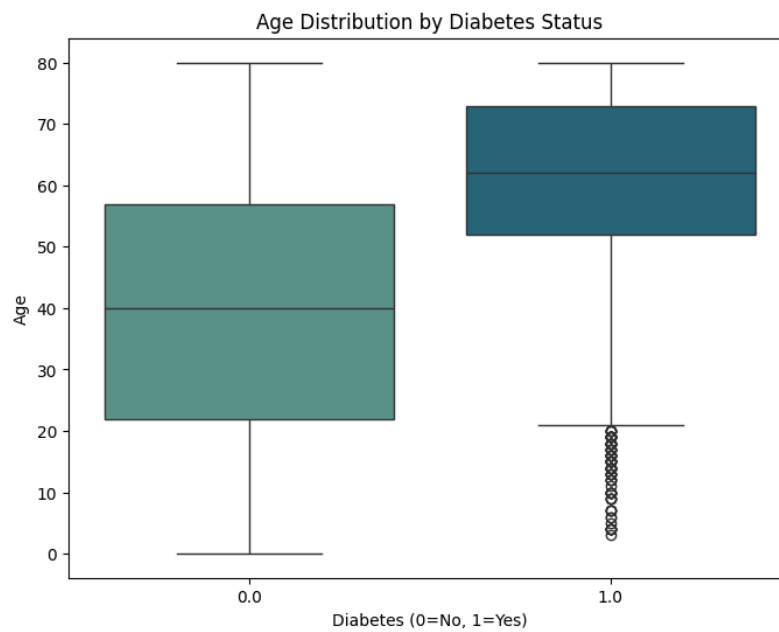


Figure 2.9: Age distribution by diabetes status.

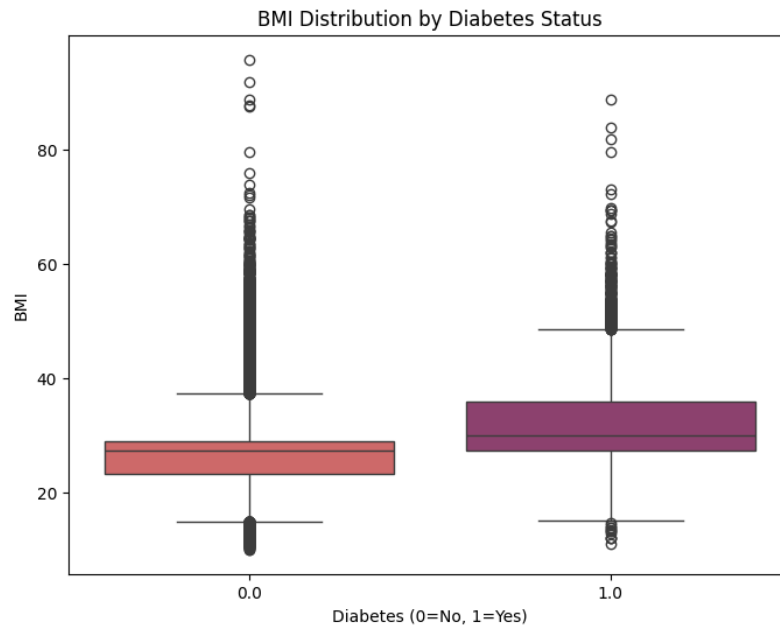


Figure 2.10: BMI distribution by diabetes status.

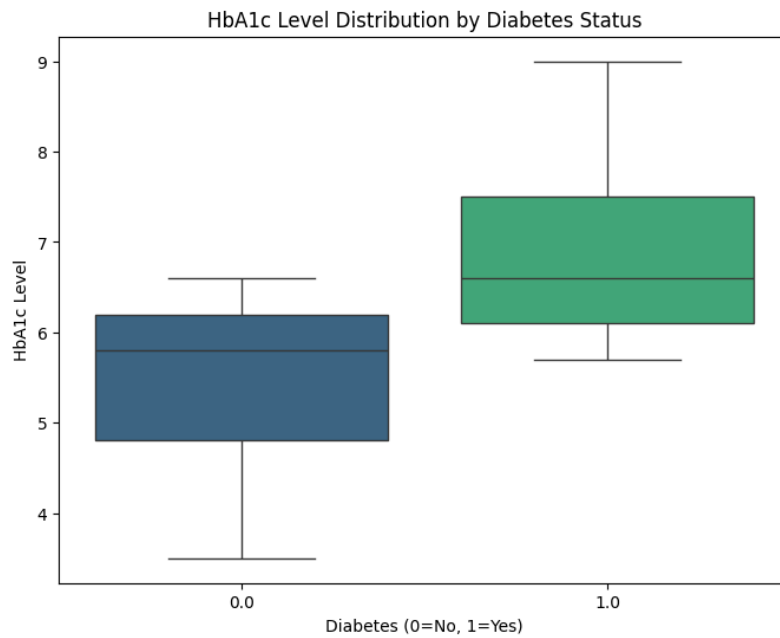


Figure 2.11: HbA1c level distribution by diabetes status.

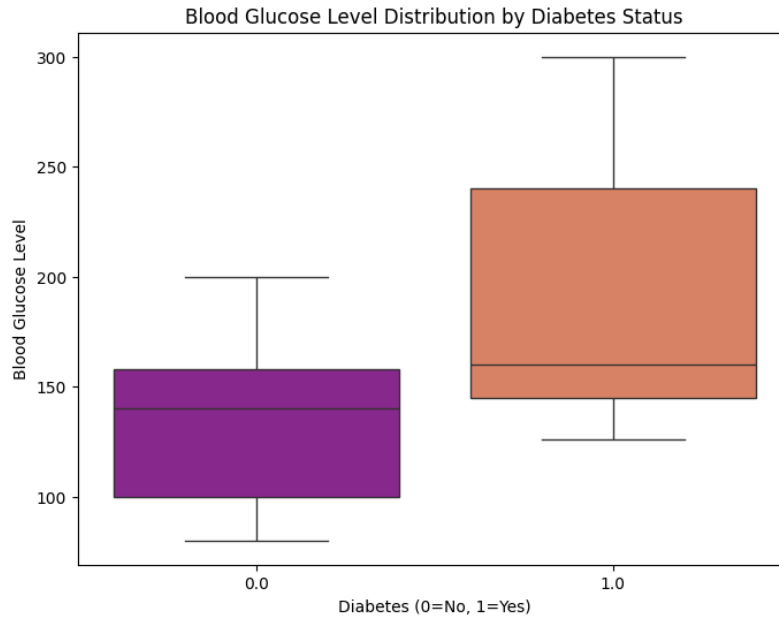


Figure 2.12: Blood glucose level distribution by diabetes status.

Overall, from the visualization, it can be seen that people who suffer from diabetes are older, have high BMI, and most importantly, have high HbA1c and high blood sugar levels compared to those who do not suffer from diabetes. The occurrence of diabetes is more frequent in those people who suffer from hypertension, cardiovascular diseases, and who smoke (past and current).

Dataset Pre-processing

Before feeding the data into any model, the dataset was analyzed for common problems in data quality and resolved them. All the problems are discussed along with the solution provided to overcome them.

3.1 Missing (Null) Values

Problem: Almost all the columns of the dataset had quite a high percentage of missing values, with 9,800 to 10,150 missing values per column out of 100,000 rows (checked using `df.isnull().sum()`).

Solution: To resolve the problem of missing values, a `SimpleImputer` from the `scikit-learn` library was used to impute missing values in different ways for numerical and categorical columns. Numerical columns like age, hypertension, heart disease, bmi, HbA1c level, blood glucose level, and diabetes were imputed with median value of each column as median is robust against outliers and skewness as compared to the mean value. On the other hand, categorical variables like gender and smoking status were filled in based on the mode of the particular column because we cannot compute an average for categorical variables.

3.2 Duplicate Rows

Problem: The dataset contained 7,866 fully duplicated rows after imputation, which could bias the model toward over-represented patient profiles and inflate apparent performance through data leakage between train and test splits.

Solution: Duplicate rows were removed with `drop_duplicates()`, reducing the dataset from 100,000 to 92,134 rows.

3.3 Columns with a Single Unique Value

Problem: Columns that contain only one unique value carry no predictive information and add unnecessary noise/dimensionality.

Solution: The dataset was checked for such columns; none were found, so no columns needed to be dropped for this reason.

3.4 Sparse Category in 'gender'

Problem: The gender column contained a small 'Other' category (only 17 records), which is too sparse to let any model learn a reliable pattern for it, and risks becoming a source of noise or overfitting.

Solution: The 17 rows labeled 'Other' were dropped, leaving a cleaner binary gender feature (Female / Male) and a final dataset shape of 92,117 rows.

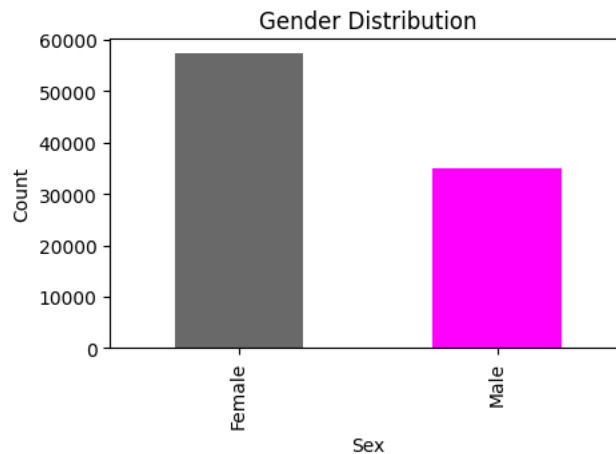


Figure 3.1: Gender distribution after removing the sparse 'Other' category.

3.5 Categorical Encoding

Problem: The `smoking_history` and `gender` columns are text-based categorical variables, but machine learning algorithms require numeric input.

Solution — `smoking_history`: Label Encoding was applied using scikit-learn's `LabelEncoder`, converting the six smoking categories (No Info, never, former, current, not current, ever) into integer codes 0–5. Label encoding was chosen here because `smoking_history` has more than two

categories, and an ordinal-style integer mapping was accepted as a practical, low-dimensional encoding for this feature.

Solution — gender: One-Hot Encoding was applied using pandas' `get_dummies()`, producing two binary columns, `gender_Female` and `gender_Male`. One-hot encoding was chosen for gender because it is a small, non-ordinal categorical variable (there is no inherent order between 'Female' and 'Male'), so one-hot encoding avoids implying a false ordinal relationship that label encoding would introduce.

3.6 Feature Scaling

Problem: The numerical features have very different scales (e.g., age ranges 0–80, `blood_glucose_level` ranges 80–300, while binary features range 0–1), which can bias distance-based algorithms such as KNN and slow down gradient-based optimization in the neural network.

Solution: Min-Max Scaling was applied using scikit-learn's `MinMaxScaler`, which was fit only on the training set and then used to transform both the training and test sets, avoiding data leakage from the test set. After scaling, every feature's minimum value became 0 and maximum value became 1 in the training set.

Dataset Splitting

The cleaned and encoded dataset (92,117 rows, with the target diabetes and the original text column `smoking_history` removed after encoding) was split into training and testing sets using a random 70/30 split via scikit-learn's `train_test_split`, with `random_state=42` for reproducibility.

Split	Rows	Percentage
Training set (X_train, y_train)	64,481	70%
Test set (X_test, y_test)	27,636	30%

A random (non-stratified) split was used, which is a reasonable approach given the very large sample size, since even with a 70/30 random split, the test set contains thousands of examples from the minority (diabetic) class.

Model Training & Testing

Four supervised models were trained on the scaled training data and evaluated on the held-out scaled test set, followed by an unsupervised K-Means clustering analysis on the full feature set.

5.1 Decision Tree Classifier

A `DecisionTreeClassifier` was trained with the Gini impurity criterion and `max_depth=10` (to control overfitting) on the training set.

Result: Test accuracy of 95.98%.

5.2 K-Nearest Neighbors (KNN)

A KNeighborsClassifier was trained using scikit-learn's default settings ($k = 5$ neighbors) on the scaled training data.

Result: Test accuracy of 95.10%.

5.3 Logistic Regression

A LogisticRegression model was trained with `max_iter=1000` to ensure convergence, on the scaled training data.

Result: Test accuracy of 94.98%.

5.4 Neural Network

A feedforward neural network was designed in PyTorch with the following structure: an input layer with 9 features, a hidden layer with 128 linear units followed by the ReLU activation function, and finally an output layer with 1 linear unit producing logits. It was trained for 1,000 epochs with the loss function as Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) and optimizer as Stochastic Gradient Descent (SGD) with a learning rate of 0.1.

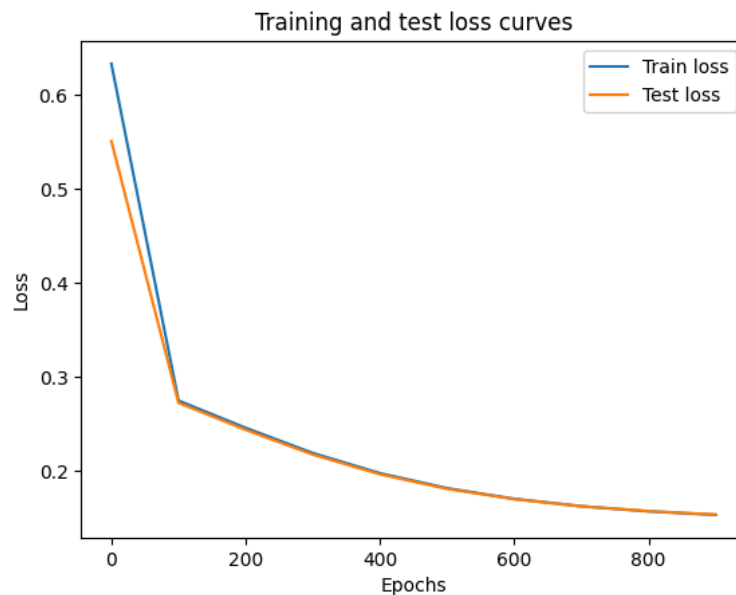


Figure 5.1: Training and test loss curves for the neural network over 1,000 epochs.

Loss plots in the training and validation show a monotonic decrease and closely match each other without any overfitting. The final accuracy achieved from the test after 1,000 epochs is around 94.4% while evaluating the test accuracy comes out to be 94.58% (refer to Section 6).

5.5 Unsupervised Learning: K-Means Clustering

To study the inherent structure of the data without any labels of diabetes, K-Means Clustering was applied to the full scaled feature set ($n = 9,117$ records) with four clusters.

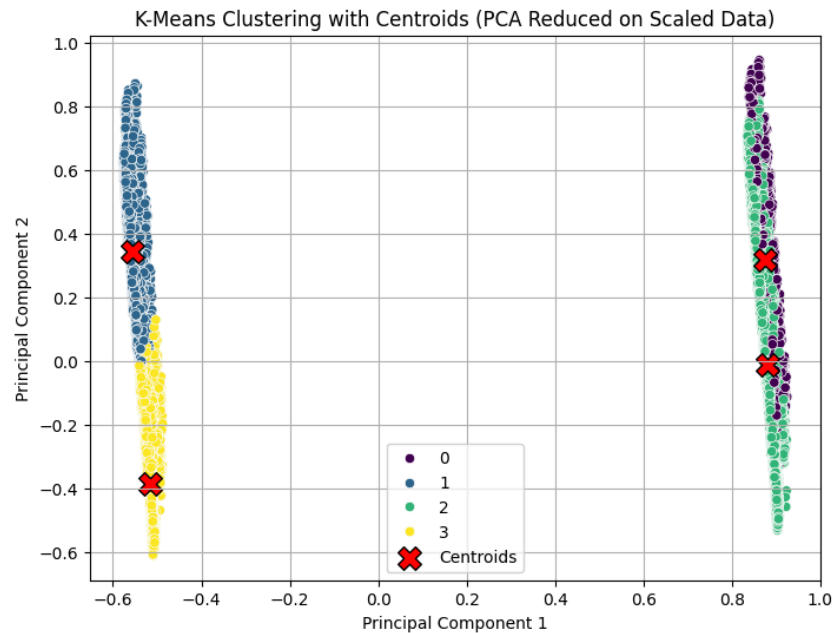


Figure 5.2: K-Means clusters ($k = 4$) visualized in PCA-reduced 2D space, with cluster centroids marked in red.

The quality of the clusters was assessed by computing the Silhouette Score. For the four clusters, the Silhouette Score came out to be 0.3956. The moderate value means that there is some amount of separation between the clusters but not perfect as some clusters may overlap.

The Elbow Method was also used, where K-Means was used to fit different numbers of clusters from 1 to 10, and the WCSS plot was created.

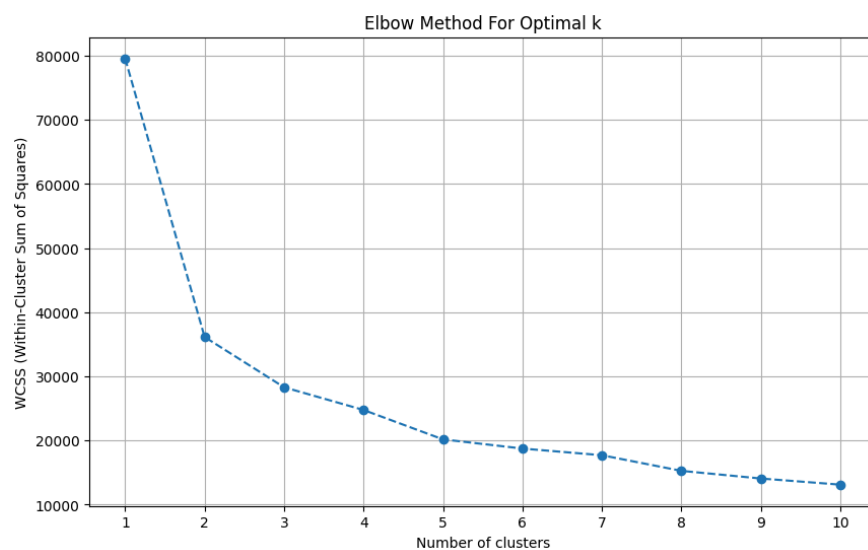


Figure 5.3: Elbow method plot showing WCSS versus number of clusters.

The within-cluster sum of squares (WCSS) plot shows a sudden drop until about 3–4 clusters, beyond which the WCSS becomes flat, implying that 3–4 clusters is a reasonable choice of this data set and is in agreement with the four clusters that have been chosen in the main analysis.

Model Selection / Comparison Analysis

The accuracy, precision, recall, F1 score, confusion matrices, and ROC-AUC measures were used to assess all the supervised algorithms on the same test set.

6.1 Accuracy Comparison

Model	Accuracy
Decision Tree	95.98%
KNN	95.10%
Logistic Regression	94.98%
Neural Network	94.58%

6.2 Precision, Recall, and F1-Score Comparison

Model	Precision	Recall	F1-Score
Decision Tree	0.8760	0.5930	0.7072
KNN	0.7996	0.5360	0.6418
Logistic Regression	0.8194	0.4971	0.6188
Neural Network	0.8424	0.4158	0.5568

While all models have achieved accuracy above 94%, recall values, i.e., the ability to detect the true diabetes cases, are notably worse for each model, varying between 41.6% and 59.3%. The mismatch between accuracy and recall is due to class imbalance mentioned in Section 2.4: with only 9% of cases being diabetic, it is possible to achieve high accuracy while missing many true positive cases. The Decision Tree has shown the best precision-recall trade-off measured by the highest F1-score of 0.7072, while the Neural Network has yielded decent precision but has demonstrated the worst recall among all four models.

6.3 Confusion Matrices and ROC Curves

Decision Tree

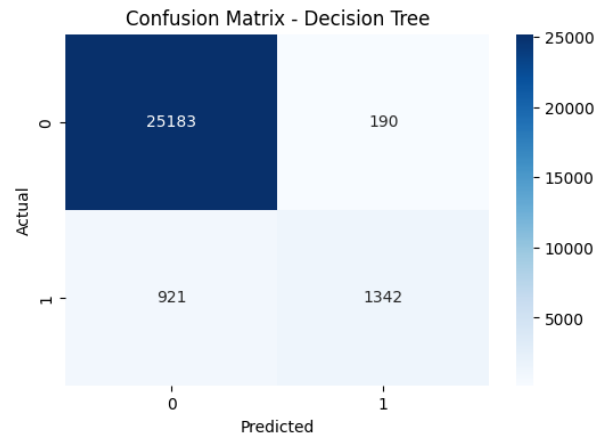


Figure 6.1: Confusion matrix — Decision Tree.

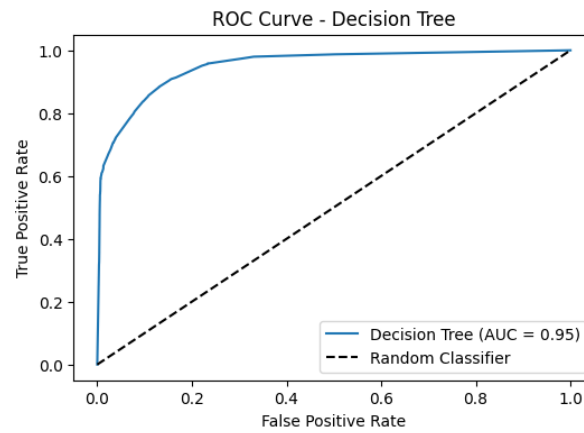


Figure 6.2: ROC curve — Decision Tree (AUC = 0.9502).

K-Nearest Neighbors

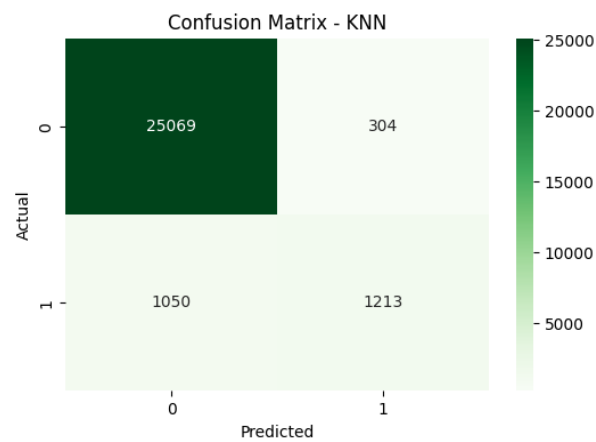


Figure 6.3: Confusion matrix — KNN.

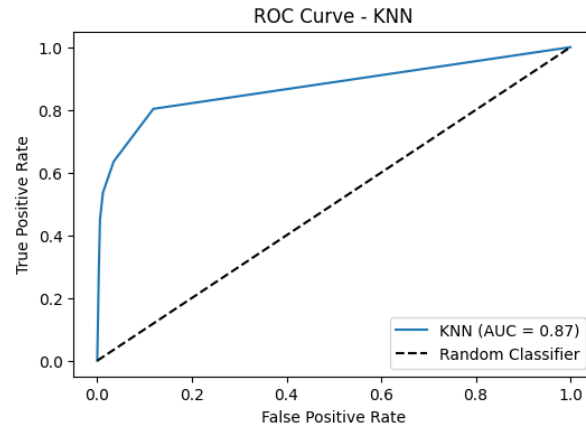


Figure 6.4: ROC curve — KNN (AUC = 0.8728).

Logistic Regression

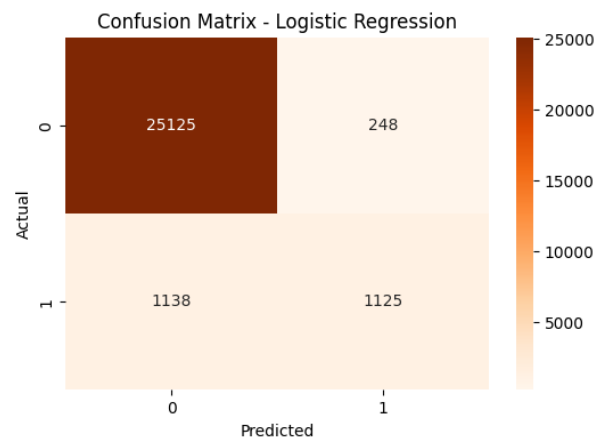


Figure 6.5: Confusion matrix — Logistic Regression.

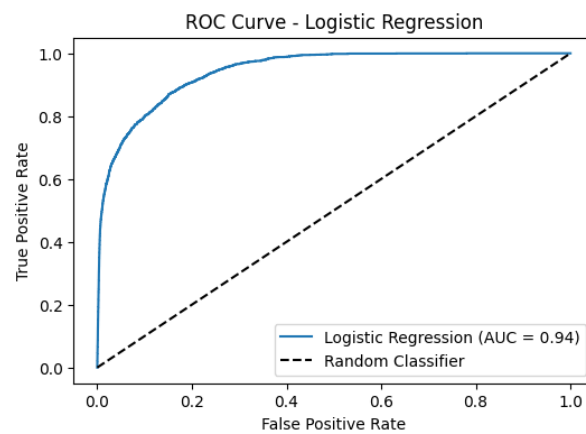


Figure 6.6: ROC curve — Logistic Regression (AUC = 0.9437).

Neural Network

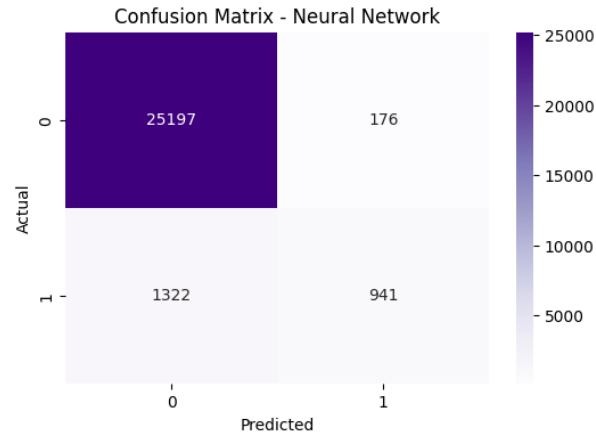


Figure 6.7: Confusion matrix — Neural Network.

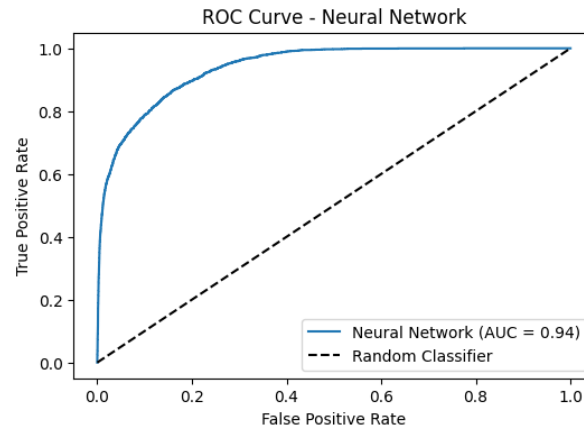
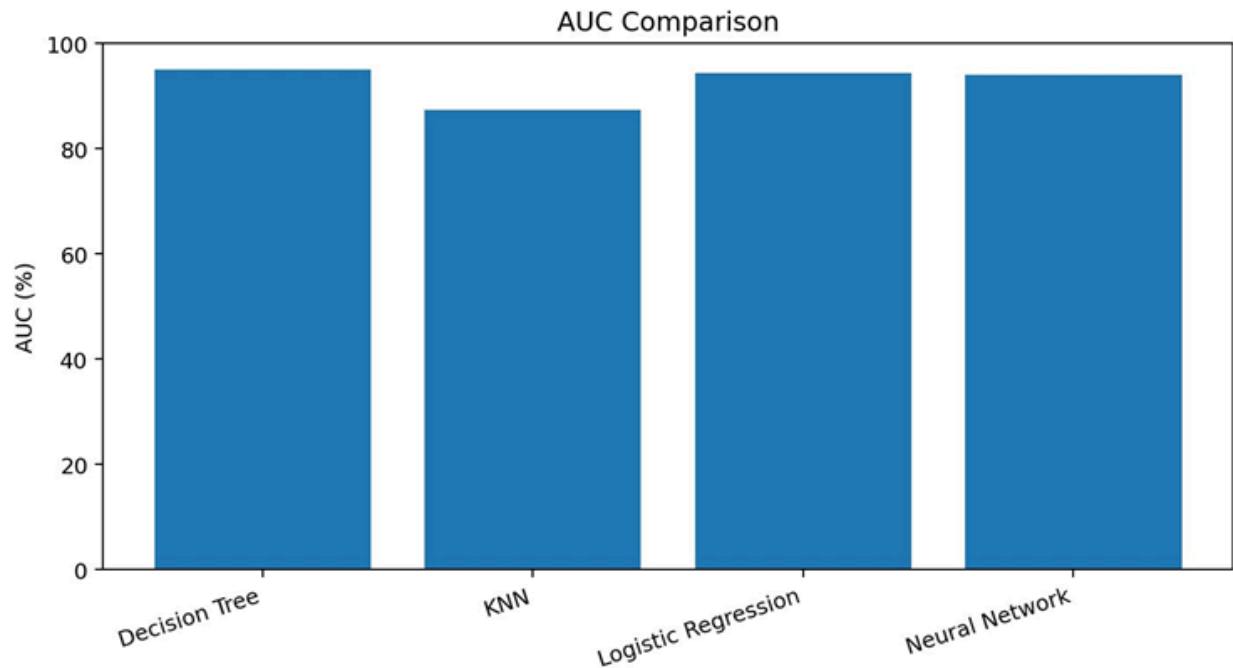


Figure 6.8: ROC curve — Neural Network (AUC = 0.9399).

6.4 AUC Score Summary

Model	AUC Score
Decision Tree	0.9502
Logistic Regression	0.9437
Neural Network	0.9399
KNN	0.8728



By the Area Under the ROC Curve (AUC) criterion, which assesses the ability of the model to differentiate between classes on all possible classification thresholds, the Decision Tree, Logistic Regression, and Neural Network demonstrate equivalent performance ($AUC \approx 0.94\text{--}0.95$), while KNN displays significant inferiority ($AUC \approx 0.87$). It means that the prediction probabilities by KNN are less reliably ranked than the probabilities estimated by the other three models, although their accuracies are roughly the same.

Conclusion

In this analysis, several machine learning algorithms are used to predict the development of diabetes using the database of 100,000 patients. Among four supervised methods, the Decision Tree was selected as the best one and showed 95.98% of accuracy, 87.60% of precision, 59.30% of recall, 70.72% of F1-score, and AUC equal to 0.9502. Nevertheless, due to class imbalance (91% of non-diabetics and 9% of diabetics), the mentioned accuracy can serve as a misleading measure, while low recall indicates that many diabetic patients have not been identified by the model. The variables that had the highest correlation with diabetes were blood glucose ($r = 0.42$) and HbA1c ($r = 0.40$). Meanwhile, the value of Silhouette Score for K-Means was equal to 0.3956. The major problems include missing values, class imbalance, categorical encoding, and clustering evaluation. Even though there were some limitations in terms of data preparation, the Decision Tree could be regarded as the best choice.